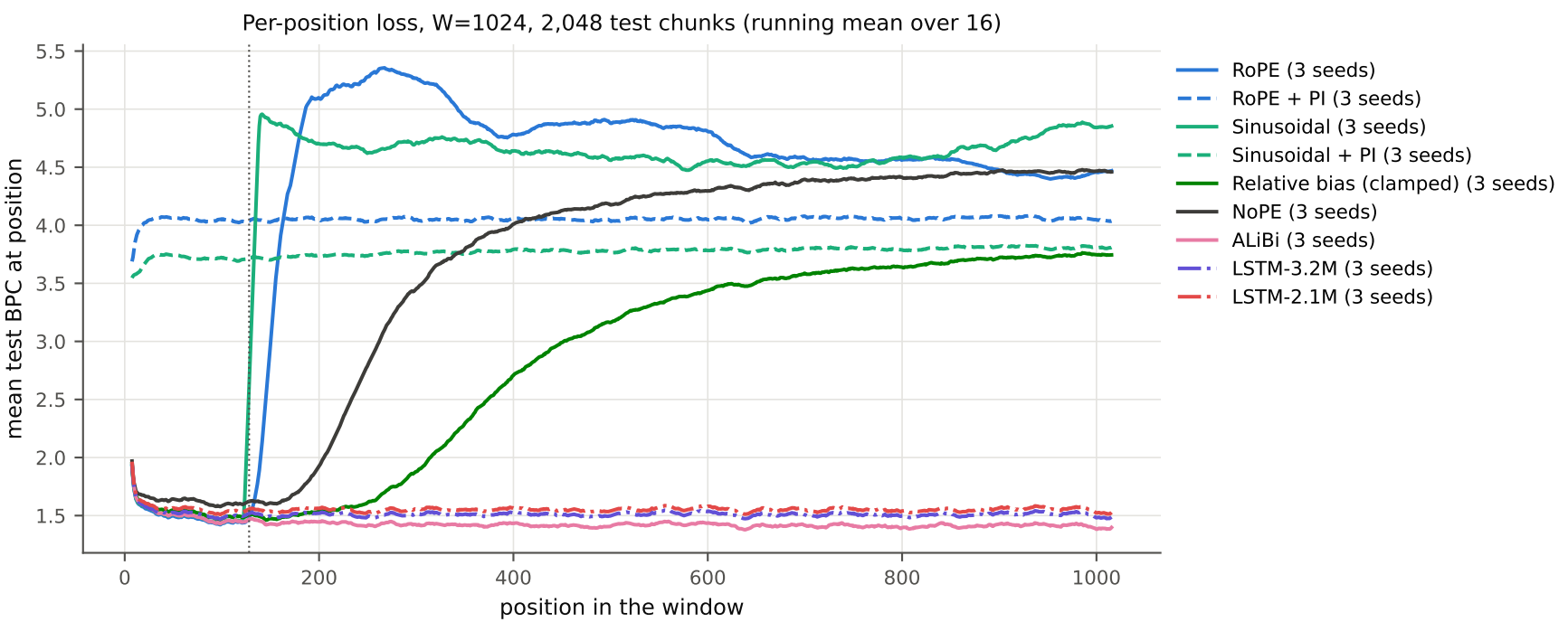

Per-position loss, W=1024, 2,048 test chunks (running mean over 16)
mean test BPC at position
position in the window
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
0
200
400
600
800
1000
RoPE (3 seeds)
RoPE + PI (3 seeds)
Sinusoidal (3 seeds)
Sinusoidal + PI (3 seeds)
Relative bias (clamped) (3 seeds)
NoPE (3 seeds)
ALiBi (3 seeds)
LSTM-3.2M (3 seeds)
LSTM-2.1M (3 seeds)